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SY3 - C3

Q1

What is binary search.

→

Binary search is a searching algorithm that repeatedly divides a sorted array into two halves. It compares the key with the middle element to find it quickly.

Q2

What is the condition for using binary search?

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The array must be sorted before applying binary search. It can be sorted in ascending or descending order.

Q3

What is the difference between iterative and recursive binary search?

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Iterative binary search uses a loop to repeat the searching process. Recursive binary search uses function calls to search the smaller halves.

Q4

What happens if the element is not in the array?

→

If  $low > high$ , the search stops because no elements are left. The function returns -1, indicating that the element was not found.

Q5

What is the time complexity of binary search?

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The best-case time complexity is  $O(1)$  when the middle element is the key. The average and worst-case complexity is  $O(\log n)$ .

Q6. Can binary search be used on unsorted arrays?

→ No, binary search requires a sorted array. For an unsorted array, we generally use linear search or sort the array first.

✓  
~~Answer~~